2.3 Training

TALKPLAY employs a principled supervised finetuning approach for LLMs, leveraging autoregressive next token prediction with cross-entropy loss. We transform synthetic conversational data (Section 3.2) by systematically replacing track identifiers with their corresponding multimodal music tokens. Formally, given a token sequence $x = \{(q_1, m_1, r_1), ..., (q_t, m_t, r_t)\}$ representing a conversation with user queries q, recommended music tokens m, and system responses r, the model is trained to predict each subsequent token in the sequence. Each token t can be either a text token from user queries or system responses, or a music token from recommended items. We use the standard autoregressive cross-entropy loss: $\mathcal{L} = -\frac{1}{n}\sum_{i=1}^{n} \log P(t_i|t_1, t_2, ..., t_{i-1})$, where $P(t_i|t_1, t_2, ..., t_{i-1})$ represents the model's predicted probability of token t_i given its preceding context.

This unified formulation enables the model to first predict contextually appropriate music tokens based on the user query $p(m|q)$, and subsequently generate coherent responses conditioned on both the query and predicted music tokens $p(r|q, m)$. Building on insights from prior research [11], we leverage the inherent coarse-to-fine structure of music modalities by predicting tokens in a carefully chosen sequence: *playlist* - *semantic* - *metadata* - *lyrics* - *audio* tokens. We validate this ordering strategy through comprehensive ablation studies on feature importance (Section 5.3).

The central goal of this approach is to effectively learn the newly introduced multimodal music token embeddings e_k^m by capturing rich contextual relationships between conversations and music items. We initialize these music token embeddings using the statistics of the base model's existing embeddings, following the vocabulary expansion method proposed by Hewitt et al. [21], $e_k^m \sim \mathcal{N}(\mu_E, \Sigma_E)$, where μ_E and Σ_E represent the mean and covariance of the base model's token embeddings.

2.4 Inference: Recommendation by Generation

During inference, we use typical LLM generation parameters (temperature=1.0, top-p=0.9, repetition penalty=1.0). The music tokens are generated before the assistant's response message to ensure recommendations are based solely on the query, avoiding potential popularity biases that could arise from conditioning on generated text [20].

The generated tokens must then be mapped back to music items through a reverse lookup of the tokenization process described in Section 2.1. While each individual cluster codebook index has a many-to-one track-to-cluster relationship, our tokenizer creates sequences of five multimodal tokens that form a nearly one-to-one track-to-cluster-sequence mapping (similar to Semantic IDs [39] using residual vector quantization), making the process more deterministic than typical approximate nearest neighbor searches used in industry recommenders [3].

Given a sequence of five multimodal tokens representing a recommendation, we select items matching the token indices across modalities. However, due to the vast token space (1,126 trillion combinations) relative to our dataset size, exact matches are often unavailable. In such cases, we fall back to partial matching using token overlap scoring with feature importance weights assigned to each modality. The final score is determined as $s_{\text{retrieval}} = \sum_{i \in \{\text{modalities}\}} \lambda_i s_i$, where each s_i term is a binary indicator (0 or 1) representing whether there is an overlap in the corresponding modality. Based on our preliminary experiment, we use a quadratically decreasing weighting scheme that best captures the relative importance of different music features. The final weights used are $\lambda_{\text{cf}} = 25$, $\lambda_{\text{semantic}} = 16$, $\lambda_{\text{metadata}} = 9$, $\lambda_{\text{lyrics}} = 4$, and $\lambda_{\text{audio}} = 1$. This weighting scheme reflects our finding that playlist co-occurrence token provide the strongest recommendation performance gain, followed by semantic tags, metadata, lyrics, and audio features. The results of these weighting experiments are presented in detail in Section 5.3.

3 Data Generation

It is prohibitively difficult to perform research on recommendation without access to real-world data – information about items, users, and their interactions. Another challenge lies in the fact that the conversational recommendation system is in its early stage and so the publicly available dataset is scarce. The Conversational Playlist Curation Dataset (CPCD) is probably the most relevant one, but it comes with 917 conversations which are not sufficient to train a large model [7].

Inspired by recent works such as [30, 12], we generate the training data using a music playlist dataset and an LLM. These synthetic datasets have been shown to closely resemble human conversations in qualitative evaluations and have proven effective for training recommendation models. Following our primary reference [12], each conversation is generated as a journey of machine-assisted music discovery; assistants continue to react by choosing music items that satisfy the user's request. A notable difference is that in TALKPLAY, the base dataset is not simply a music dataset (e.g., [4]) but a playlist dataset, in order to simulate more plausible and coherent conversations and recommendations.

3.1 Base Dataset: Million Playlist Dataset

In the current era of music streaming, playlists are playing a crucial role in the music listening experience, providing unique and direct information about music items: which music items go along together well, according to the real listeners. They may suit well for the purpose of training TALKPLAY as well, as a base dataset, since conversation consisting of tracks in the same playlist would be somewhat diverse yet coherent, and hence realistic.

Table 1: Statistics of the TALKPLAY dataset.

Dataset	Training	Evaluation
Million Playlist Dataset (MPD)		
# of Playlists	999,000	1,000
# of Average Tracklist	63.73	59.41
# of Warm Tracks	1,714,772	37,114
# of Cold Tracks	-	2,357
# of Vocab	24,129	713
TalkPlay Dataset		
# of Playlist	116,099	1,000
# of Dialogue	531,552	1,000
# of Tracks	405,543	11,006
# of Vocab	317,705	19,736
Avg. # of turns	6.95	13.05

The Million Playlist Dataset (MPD) is chosen as our base dataset [8]. Since being introduced in 2018, this dataset remains one of the largest playlist datasets available to the public by including one million Spotify playlists. The MPD provides track metadata and playlist co-occurrence and is extended with audio content, lyrics, and semantic annotations. The audio content are successfully crawled through the Spotify API for 1,717,148 items. The lyrics are extracted using WHISPER-LARGE-V3 [38], a well-known speech transcription model that performs decent lyrics transcription. For semantic annotations, we leverage pretrained music understanding models to extract comprehensive musical attributes. Specifically, we use LP-MUSICCAPS [13] to extract high-level semantic features including genres, moods, instruments, and vocal styles. Additionally, we employ MADMOM [5] to extract structural music properties such as key signatures and tempo information. This multi-faceted approach ensures our model has access to a rich representation of musical characteristics.

3.2 TalkPlay Dataset

We use an LLM and the base dataset (MPD) to generate TalkPlay Dataset, a music conversation dataset with natural language and music recommendation.[3] These are the key design principles and choices to ensure the synthetic data can be used for the training of TALKPLAY.

Coherence: Given a user query/reaction, system should recommend music with relevant dialogue.

Modality Coverage: Conversations span four target modalities: audio, lyrics, metadata, and semantic tags. The playlist co-occurrence is excluded because: i) this modality is implicit and difficult to verbalize, and ii) in our task setup, playlist co-occurrence would be inherently captured (Section 2.1).

User Simulation: The user may skip or reject the recommended tracks, requesting for alternative recommendations. For rejected tracks, we employ hard negative sampling by using songs from artists whose tracks appear in the playlist. This approach ensures that the rejected recommendations are plausible but not suitable for the current conversation context.

Text Format: The LLM output, the synthesized conversation, should be in a structured format. The JSON format is chosen and used as presented in Section B.2

Based on this, GEMINI-1.5-FLASH-002 [44] is used to create music conversation given a prompt that includes i) our goal and the aforementioned conditions, ii) track identifiers and their metadata, semantic tags, lyrics. The prompt is included in Section B.1. Table 1 compares our dataset with the base dataset. Using 116k playlists with an average of 7 tracks each, our synthesis approach yields 532k multi-turn music recommendation dialogues. By leveraging rich textual annotations

[3] We will release the dataset to the public after the review period.

of music sequences, our LLM-based data synthesis technique significantly expands the text-music associations beyond just playlist titles, increasing the token count from 24k to 318k. Each user-assistant interaction in our synthetic conversation dataset follows a triplet structure: {USER QUERY, RECOMMENDED MUSIC, ASSISTANT RESPONSE}. The user query and assistant response are natural language expressions, while the recommended music is represented by a unique track identifier.

Data Split: To reflect the real-world recommendation scenarios, we perform a chronological data split based on the playlist creation date. The test set is created by randomly sampling 1,000 playlists from the last creation date (2017-11-01), and training set include all the rest of the tracks. Through this split, around 2,357 tracks in the test split are cold-start items, tracks that only appear in the test set. Assuming a real-world scenario, these items represent newly released songs, which traditional collaborative filtering algorithms have no information to make recommendation from. In contrast, being multimodal, TALKPLAY is able to handle these items by understanding content such as audio, lyrics, metadata and semantic features.

4 Experiments

We conduct a comprehensive evaluation of TALKPLAY, focusing on two critical dimensions: (1) retrieval performance - the model's ability to recommend relevant music items based on multi-turn dialogue context, and (2) response quality - the model's capability to generate contextually appropriate, natural language responses that explain recommendations. Through both objective metrics and human evaluation, we examine how our generative approach compares to traditional retrieval methods across various conversation scenarios and user preferences.

Objective Evaluation: For the multi-turn recommendation evaluation, we simulate a turn-taking scenario where each turn is conditioned on the previous turns and the current turn's user query, with the goal of retrieving appropriate music recommendations. Following previous works [7, 12], we employ standard ranking metrics - Mean Reciprocal Rank (MRR) and Hit Rate at K (Hit@K). We evaluate retrieval performance for each turn using 1,000 multi-turn conversations derived from 1,000 playlists in Table 1.

Subjective Evaluation: To assess the real-world usability of TALKPLAY, we conducted rigorous A-vs-B human preference evaluations against baseline models and ground truth. In each evaluation scenario, evaluators compared a pair of outputs (recommendations and responses) from different models based on a shared conversation context, to choose a preferred model (or a tie) on the two critical dimensions: (1) *Recommendation Relevance* - how well the recommended music items aligned with the conversation context, and (2) *Response Naturalness* - the conversational quality and contextual appropriateness of the generated text responses. We recruited 18 evaluators with expertise in music research or professional experience in the music industry. Each evaluator assessed 10 randomly selected conversation scenarios, yielding 180 total ratings.

Training Details: Using the synthetic conversation dataset (Section 3), TALKPLAY, a vocabulary-expanded Llama-3.2-1B model, is trained with a 1024-token context length, a learning rate of $1e-4$, and the AdamW optimizer [32]. The model reported in this paper is trained on 8 GPUs with a total batch size of 48 items (or 49,152 tokens).

Baseline Models: We evaluate our model against three baseline approaches: BM25 [40] for classical text-based sparse retrieval, NV-Embeds-V2 [29] for deep similarity retrieval with LLM embeddings, and CLAP [49] for multimodal audio-text retrieval. We also include two generative retrieval models, SASRec [26] and TIGER [39]. While both models perform next token prediction for recommendation, SASRec represents each of the 400k tracks as unique IDs, while TIGER uses Semantic IDs that combine SentenceT5 [35] embeddings with residual vector quantization (RVQ) for track representation. Unlike TALKPLAY, SASRec and TIGER lack natural language understanding capabilities. As a result, these models can only make recommendations based on track sequences, ignoring the semantic content of user queries in multi-turn conversations. Additional details can be found in Appendix D.

For our subjective evaluation, we chose BM25 [40] and NV-Embeds [29] as baseline models, as they demonstrated strong query understanding capabilities in our objective evaluation (In Table 2). To enable these retrieval-based models to generate conversational responses, we implemented a two-stage CRS using Llama-3.2 1B [16]. This pipeline first retrieves relevant music items using retrieval model, then uses these retrieved items as context for the LLM to generate appropriate responses.